

Patient-managed interventions for adults with bronchiectasis: evidence, challenges and prospects.

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Bronchiectasis is a chronic lung condition which is characterised by recurrent chest infections, chronic sputum production and cough, and limited exercise tolerance. While bronchiectasis may be caused by various aetiologies, these features are shared by most patients with bronchiectasis regardless of the cause. This review consolidates the existing evidence on patient-managed interventions for adults with bronchiectasis, while also outlining areas for future research. Airway clearance techniques and hyperosmolar agents are key components of the bronchiectasis management and consistently recommended for clinical implementation. Questions around their prescription, such as optimal sequence of delivery, are still to be answered. Pulmonary rehabilitation and exercise are also recommended for patients with bronchiectasis. Relatively strong evidence underpins this recommendation during a clinically stable stage of the disease, although the role of pulmonary rehabilitation following an exacerbation is still unclear. Additionally, self-management programmes feature prominently in bronchiectasis treatment, yet the lack of consensus regarding their definition and outcomes presents hurdles to establishing a cohesive evidence base. Moreover, cough, a cardinal symptom of bronchiectasis, warrants closer examination. Although managing cough in bronchiectasis may initially appear risky, further research is necessary to ascertain whether strategies employed in other respiratory conditions can be safely and effectively adapted to bronchiectasis, particularly through identifying patient responder populations and criteria where cough may not enhance airway clearance efficacy and its control is needed. Overall, there is a growing recognition of the importance of patient-managed interventions in the bronchiectasis management. Efforts to improve research methodologies and increase research funding are needed to further advance our understanding of these interventions, and their role in optimising patient care and outcomes.

Risk of Severe Bleeding With Extended Rivaroxaban to Prevent Venous Thromboembolism in Acute Medically Ill Patients With Bronchiectasis.

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Background: Bronchiectasis is a chronic inflammation of the bronchi with recurrent infections and hemoptysis. The MAGELLAN study compared oral rivaroxaban, 10 mg once daily (QD), for 35 ± 4 days with subcutaneous enoxaparin 40 mg QD for 10 ± 4 days followed by placebo for 25 ± 4 days to prevent venous thromboembolism in patients hospitalized with an acute medical illness. MAGELLAN included a subset of patients with bronchiectasis. In a post hoc analysis, we evaluated the incidence and severity of pulmonary bleeding in patients with bronchiectasis who were hospitalized for an acute medical illness. This analysis included MAGELLAN patients diagnosed with bronchiectasis at baseline. Patients were evaluated by treatment group for International Society on Thrombosis and Haemostasis major bleeding, non-major clinically relevant (NMCR) bleeding, and the composite of the 2 (ie, clinically relevant bleeding). Results: Medically ill patients with bronchiectasis were randomized to rivaroxaban ($n = 60$) or enoxaparin/placebo ($n = 61$). There were 2 fatal pulmonary bleeds and 1 fatal gastrointestinal bleed in the rivaroxaban arm and no fatal or major bleeding in the enoxaparin/placebo arm. The incidence of major bleeding was 5% in the rivaroxaban arm. One NMCR bleed occurred in the rivaroxaban arm and 2 NMCR bleeds occurred in the enoxaparin/placebo arm. The incidence of clinically relevant bleeding was 6.7% versus 3.3% in the rivaroxaban and enoxaparin/placebo groups, respectively (relative risk = 2.06

[95% confidence interval: 0.351-12.046]). Conclusion: In-patients hospitalized with bronchiectasis and an acute medical illness, clinically relevant bleeding, including fatal pulmonary hemorrhage, occurs more frequently with extended rivaroxaban thromboprophylaxis than with enoxaparin followed by placebo.

Viruses in bronchiectasis: a pilot study to explore the presence of community acquired respiratory viruses in stable patients and during acute exacerbations.

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Bronchiectasis is a chronic respiratory condition. Persistent bacterial colonisation in the stable state with increased and sometimes altered bacterial burden during exacerbations are accepted as key features in the pathophysiology. The extent to which respiratory viruses are present during stable periods and in exacerbations is less well understood. This study aimed to determine the incidence of respiratory viruses within a cohort of bronchiectasis patients with acute exacerbations at a teaching hospital and, separately, in a group of patients with stable bronchiectasis. In the group of stable patients, a panel of respiratory viruses were assayed for using real time quantitative PCR in respiratory secretions and exhaled breath. The Impact of virus detection on exacerbation rates and development of symptomatic infection was evaluated. Routine hospital-based viral PCR testing was only requested in 28% of admissions for an exacerbation. In our cohort of stable bronchiectasis patients, viruses were detected in 92% of patients during the winter season, and 33% of patients during the summer season. In the 2-month follow up period, 2 of 27 patients presented with an exacerbation. This pilot study demonstrated that respiratory viruses are commonly detected in patients with stable bronchiectasis. They are frequently detected during asymptomatic viral periods, and multiple viruses are often present concurrently.

Bronchiectasis in adults: a review.

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Bronchiectasis is characterized by permanent, abnormal dilation of the bronchi with chronic inflammation and infection. Bronchiectasis is either caused by or associated with several systemic disorders, and in a significant number of cases, the underlying etiology remains unidentified and is subsequently referred to as idiopathic. The clinical course is characterized by chronic cough with purulent sputum associated with acute exacerbations. Patients may develop progressive decline in pulmonary function due to ongoing infection, inflammation, and destruction of bronchi if a delay in diagnosis and therapy occurs. With high-resolution computed tomography, the confirmatory diagnosis of bronchiectasis can be made promptly and diagnostic work-up for etiology initiated. Identifying a specific etiology, when present, is critical for implementing appropriate treatment of the underlying cause. Whether a specific etiology can be identified or not, all cases of bronchiectasis require management that addresses infection, inflammation, and bronchial toilet.

Acute and long-term management of severe bronchiectasis with high flow nasal therapy: A case report.

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Bronchiectasis (BE) is a long-term, chronic lung condition featured by widened and scarred airways. These can alter the physiological mucociliary clearance, making it difficult to clear mucus and microorganisms, leading to frequent exacerbations. High flow nasal therapy (HFNT) is a noninvasive respiratory support that delivers heated and humidified gas eventually enriched with oxygen, through a nasal cannula. Humidification is crucial for adequate airways mucociliary clearance, improving ciliary function and consequently reducing airways inflammation and recurrent infections. HFNT has been mostly used in patients with acute hypoxemic respiratory failure and in selected patients with chronic respiratory failure due to COPD. Still, evidence about its use in acute and long-term home setting in patients with clinically relevant BE are lacking. We report a case of severe widespread BE, already on top medical therapy and pulmonary rehabilitation, still suffering from difficult mucus expectoration and recurrent exacerbations, who has been additionally treated with HFNT, both in hospital and domiciliary, reporting significant improvements on relevant clinical and patient-centered outcomes. Thus, HFNT may confer additional benefits as an add-on treatment of patients with severe BE and respiratory failure.

Severe bronchiectasis.

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Bronchiectasis is primarily the result of airway injury and remodeling attributable to recurrent or chronic inflammation and infection. The underlying etiologies include autoimmune diseases, severe infections, genetic abnormalities, and acquired disorders. Recurrent airway inflammation and infection may also be the result of allergic or immunodeficiency states such as allergic bronchopulmonary mycoses or HIV/AIDS. Bronchiectasis should be included in the differentiation diagnosis of any patient with chronic respiratory complaints such as cough and sputum production. Early clinical manifestations may be subtle. Hallmarks of severe bronchiectasis include fetid breath, chronic cough, and sputum production. The associated chronic respiratory infections and airway sepsis are punctuated by episodes of acute exacerbation. Prompt recognition and treatment of bronchiectasis may allow for prevention of disease progression and irreversible loss of lung function. This review of severe non-cystic fibrosis bronchiectasis describes the current pathophysiology, clinical presentations, and management of bronchiectasis. We review how impaired airway clearance and the inability to resolve infection and inflammation creates a vicious cycle of recurrent injury. The common clinical features of bronchiectasis and findings are presented and illustrated by radiographic images. The common species and significance of various organisms often recovered from the distal airways including: tuberculous and environmental mycobacteria, aspergillus, and bacteria such as *Pseudomonas aeruginosa* will be covered. Management strategies including sputum surveillance, sputum clearance, antimicrobial therapy including antifungal and antimyobacterial agents as well as the evidence for the use of inhalational and anti-inflammatory therapies such as corticosteroids are also discussed. Recommendations for the work-up and therapy of complications including hemoptysis and respiratory failure are presented.

Effects of Pulmonary Rehabilitation on Systemic Inflammation and Exercise Capacity in Bronchiectasis: A Randomized Controlled Trial.

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Bronchiectasis is a chronic condition that is becoming a global health concern. To examine the effects of pulmonary rehabilitation (PR) on systemic inflammation, exercise capacity, and quality of life in participants with bronchiectasis. Participants were randomized to receive PR (outpatient, three weekly sessions for 3 months) or control intervention (usual care + airway clearance therapy + breathing exercises). Data on laboratory (fibrinogen level) and patient-centered outcomes such as physical fitness [6-min walk test (6MWT)] and quality of life were collected. A total of 41 participants were evaluated (20 in the intervention group and 21 in the control group). The magnitude of change between baseline and the end of study was greater in the PR group than in the control group-the 6MWT distance increased by a mean of 54 m (54 vs 12 m; $p < 0.01$), fibrinogen showed a significant reduction (fibrinogen - 92.8 versus - 47.1 mg/dl; $p < 0.01$), and quality of life improved according to Saint George's Respiratory Questionnaire (SGRQ) (- 7.5 vs 3.2; $p < 0.01$), which exceeded the minimal clinically important difference of 4 points. PR effectively improved physical fitness, quality of life, and the degree of systemic inflammation, as reflected by changes in 6 MWT, fibrinogen levels and SGRQ scores. This study supports the inclusion of people with bronchiectasis in supervised PR programs.

The clinical phenotype of bronchiectasis and its clinical guiding implications.

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Bronchiectasis is a chronic airway disease with abnormal and persistent bronchial dilatation caused by a variety of reasons. In recent years, numerous reports have shown that bronchiectasis is heterogeneous, the clinical characteristics of patients with different phenotypes are different, and the efficacy of a treatment regimen may vary greatly in patients with different bronchiectasis phenotypes. This paper summarizes the current clinical phenotypic classification of bronchiectasis from the perspective of etiology, microbiology, and the frequency of acute exacerbation, and cluster analysis was used to determine new clinical phenotypes and their statistical and clinical significance. Different tools for assessing disease severity yield different outcomes. This article summarizes the research progress in the above areas, hoping to provide a more comprehensive understanding of the disease.

Diagnosis and management of non-cystic fibrosis bronchiectasis.

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Bronchiectasis is a heterogeneous and increasingly prevalent chronic pulmonary disease that is associated with significant morbidity. In this review, we outline how patients with bronchiectasis may present clinically and describe an approach to its diagnosis, including how to identify an underlying aetiology. We discuss the important considerations when treating either acute exacerbations or stable disease and provide an overview of the role of long-term antimicrobials, airway clearance methods and other supportive management.

A practical guide to the diagnosis and management of suspected Non-tuberculous Mycobacterial Pulmonary Disease (NTM-PD) in the United Kingdom.

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Non-tuberculous Mycobacterial Pulmonary Disease (NTM-PD) is a chronic disease characterised by progressive inflammatory lung damage due to infection by non-tuberculous mycobacteria (NTM). Global prevalence of NTM-PD is generally low but is rising, likely due to a combination of increased surveillance, increasing multimorbidity and improved diagnostic techniques. Most disease is caused by Mycobacterium avium complex species. NTM-PD can be challenging to both diagnose and manage but given the risk of untreated disease and the challenges around drug treatments, it is vital that all healthcare professionals involved in primary care consider NTM-PD at the earliest opportunity. In particular, NTM-PD should be considered where there are respiratory symptoms in the setting of pre-existing chronic lung disease such as chronic obstructive pulmonary disease (COPD) and bronchiectasis. Early suspicion should lead to appropriate primary screening measures. This article discusses the relevance of NTM-PD today, risk factors for developing disease, pathways from clinical presentation to referral to specialist care, and discusses management and drug treatments. A flow diagram of a screening process is presented as a guideline for best practice from a United Kingdom perspective.
